

Review of Algebraic Expressions and Equations

A variable represents an unknown quantity.

A term is a number, variable or a product of numbers and variables.

$$9$$

$$6ab$$

$$y$$

Algebraic expressions are made up of terms.

$8x$ is an algebraic expression with one term

$x + 5$ has two terms: x and $+5$

$5a - 3b$ also has two terms: $5a$ and $-3b$

Terms with the **same variables raised to the same exponents** are called like terms. Otherwise, they are unlike terms.

Like Terms	Unlike Terms	
$3x, 2x$	$5x, 5x^2$	Same variable x , but different exponents
$-6x^2y, 2x^2y, 4x^2y$	$7y, 3z, 8x$	Different variables
$2ab^2c^3, ac^3b^2$	$6abc^3, 6ab^2$	Different variables and different exponents

Determine whether the pairs of terms are like terms.

a) $2x^2y$ and $3x^2y$

b) $4x^2y^2$ and $7x^2y^3$

c) $3xyz^4$ and $3x^4yz$

d) $17x^3y^4z^5$ and $8x^3y^4z^5$

Only like terms of algebraic expressions can be combined when adding or subtracting. When given an expression, you may rearrange the terms so that like terms are next to one another.

Simplify.

a) $3x + 2x$

b) $5x^2 - x^2$

c) $2x^2 + 5x - 3x^2 - x$

d) $4ab + 3bc - 5ab + 5b^2 - 2bc$

e) $2xy + 4x^2y - 6xy^2 + 8x^2y$

To multiply an algebraic expression by a real number,

- Using the distributive property, multiply the coefficients but leave the variables unchanged.
- Simplify by combining like terms.

Multiply and simplify.

a) $6(a + b)$

b) $-3(2x - y - 5)$

c) $-(a + 4b - 2c)$

d) $2(x + y) - 5(x - y)$

Add and subtract polynomials

To add polynomials, combine any like terms.

To subtract polynomials, we must be careful with the minus sign in front of a set of parentheses. Think of the minus sign as being negative. We must distribute it which will change the signs of all terms inside the parentheses. Then, combine any like terms.

Note:

Most students do not make mistakes with the variables or their exponents. The problem tends to be with the coefficients and their signs.

Simplify.

a) $(7x^2 - 6x + 3) - (5x^2 - 8x - 12)$

b) $(2x^2 - 4x + 3) + (5x^2 - 6x + 1)$

c) $(-6x^2y - 3xy + 7xy^2) - (5x^2y - 8xy - 5x^2y^2)$

Equations

An **equation** uses an equals sign (=) and indicates that two expressions are equal. An **algebraic equation** is an equation that contains at least one variable.

A linear equation is one that takes the form $ax + b = c$ (where a , b , and c are constants)

For example, $x + 1 = 9$ and $x - 5 = 2$ are linear equations.

The Addition (Subtraction) Principle

Equal quantities may be **added to (or subtracted from) both sides** of an equation without changing the solution or equality.

Solve for x .

a) $x - 15 = -5$

b) $14 = x - 3$

c) $1.5 + 0.2 = 0.3 + x + 0.2$

The Multiplication Principle

Both sides of an equation may be **multiplied** by the **same quantity** (except zero).

Both sides of an equation may be **divided** by the **same quantity** (except zero).

Note: When the coefficient of x is a fraction, we can get x alone by multiplying by the reciprocal of the coefficient.

Solve for x .

a) $2x = 20$

b) $\frac{x}{3} = 10$

c) $-3x = 48$

d) $\frac{5}{2}x = 15$

To solve linear equations of the form $ax + b = c$:

- Use the addition principle to obtain an equation that has the term containing the variable on one side and the constants on the other side.
- Use the division or multiplication principle to solve for the variable.

Solve.

a) $3x + 2 = 17$

b) $5x + 3 = -27$

c) $9 - 4x = 13$

d) $16 = 12 - 5x$

To solve equations with a variable on both sides of the equation:

- Combine like terms on each side of the equation.
- Use the addition principle to obtain an equation that has the terms containing the variable on one side of the equals sign and the constants (numbers) on the other side.
- Use the multiplication principle to solve for the variable.

Solve.

a) $5x + 3 = 3x - 17$

b) $2x - 5 = 5x + 7$

c) $0.6y + 0.8 = 0.1 - 0.1y$

To solve linear equations with parentheses:

- Use the distributive property to remove the parentheses
- Proceed as you did for the previous examples.
 - Combine like terms on each side of the equation.
 - Use the addition principle to obtain an equation that has the terms containing the variable on one side of the equals sign and the constants (numbers) on the other side.
 - Use the multiplication principle to solve for the variable.

Solve.

a) $4(x + 1) - 3(x - 3) = 2$

b) $2(3x - 5) + 2 = 4x - 3(8 + 2x)$

c) $10 + 3(1 + 2x) = 5[2x - (1 - x)]$

1.6 Solve problems using inequalities

We often speak of one value as being greater than or less than another value. These relationships are called inequalities.

“6 is less than 12.”

“12 is greater than 6.”

These statements can also be written using mathematical symbols.

$>$ represents the words “is greater than”

$<$ represents the words “is less than”

Inequality	Symbol	Visual	Description
Greater than	$>$		All points to the right of the open circle are included in the solution set.
Greater than or equal to	$\geq$		The circle and all points to the right of the circle are included in the solution set.
Less than	$<$		All points to the left of the open circle are included in the solution set.
Less than or equal to	$\leq$		The circle and all points to the left of the circle are included in the solution set.

Inequalities may be solved in the same way we solve equations. However, if you multiply or divide by a negative number, the direction of the inequality must be reversed.

Note: To avoid any confusion, it is recommended that you move your variables to the left of the inequality when solving.

Solve the inequalities.

a) $3x + 7 \geq 13$

b) $5x - 2 > 18$

c) $3x - 8 > 7x + 4$

d) $2(3x - 2) \geq 3(1 - x) + 2$

Determine whether the inequalities are true or false when $A = 1$, $B = -2$, and $C = 0$.

a) $A \geq B$

b) $-A < C$

c) $2A > -B$

d) $2A + B \leq 0$

1.7 Solve problems requiring the transposition of equations

We can rearrange formulas using the same method that was previously used to solve equations. Our goal is to isolate a certain variable so we will have to “get rid of” everything else.

Example:

If $D = L - N$, solve for L .

To solve for L , we must isolate L . In other words, we need $L = \text{something}$. To get L by itself, we must add N to both sides of the equation.

$$D + N = L - N + N$$

$$D + N = L$$

Solve for the indicated variable.

a) $4p - 3t = 1$ Solve for p .

b) $PV = nT + R$ Solve for T .

c) $A = \frac{1}{2}bh$

Solve for h.

d) $P = 2L + 2W$

Solve for W.

e) $F = \frac{9}{5}C + 32$

Solve for C.